

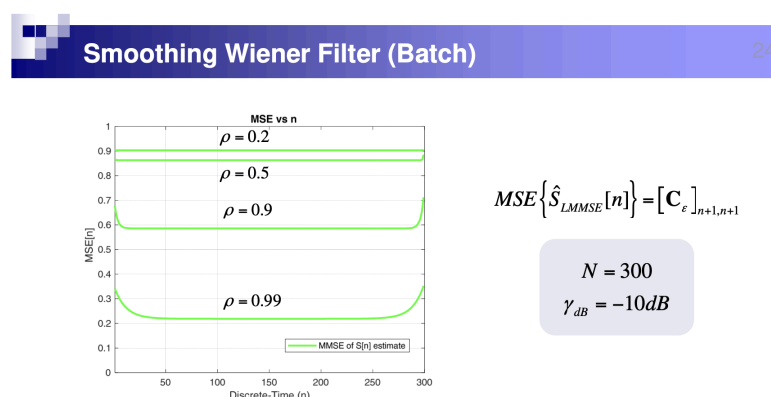
DISCLAIMER: If you don't like the total vote, you probably have to do another time, also the written part.

FIRST QUESTION:

1. Asked about the large number law and how it is connected to the sample estimator (all the demonstration, from the Binary random variable to the frequency) . Also what's the condition for the estimator to be consistent ($\text{var} < +\infty$) (Third pdf from 23-32, 41-48 and 56-65).
2. Talk about the Least Squares estimator, from the general definition (slide 4 chapter 5) to the geometric interpretation in the scalar case (slides 10-23 chapter 5). Explain the differences between LS and ML. Apply the least squares method to the estimation of the amplitude of a signal with known shape (example from slides 47–48, Chapter 5).
3. Linear prediction
4. Describe an ARMA model and how to derive its coefficient (do all the demonstration and write all what you remember)
5. LMMSE estimator: its definition and geometrical interpretation, the derivation of the formula in the scalar case (regression line), and the writing of the formula in the vectorial case. In general pros and cons and when it is used.
6. [written one] Talk about LMMSE, focus on differences with MMSE, geometrical interpretation, when it coincides with MMSE

SECOND QUESTION

1. Talk about this graph:



- Denote by L_s is the correlation time of the process $S[n]$.
- The estimation accuracy of the first L_s and of the last L_s signal samples (i.e. at the beginning and at the end of the observation interval) is lower than for the other samples.

2. In the context of the Wiener smoothing filter, suppose we observe N samples of a process $x[n] = s[n] + w[n]$, with s and w uncorrelated.

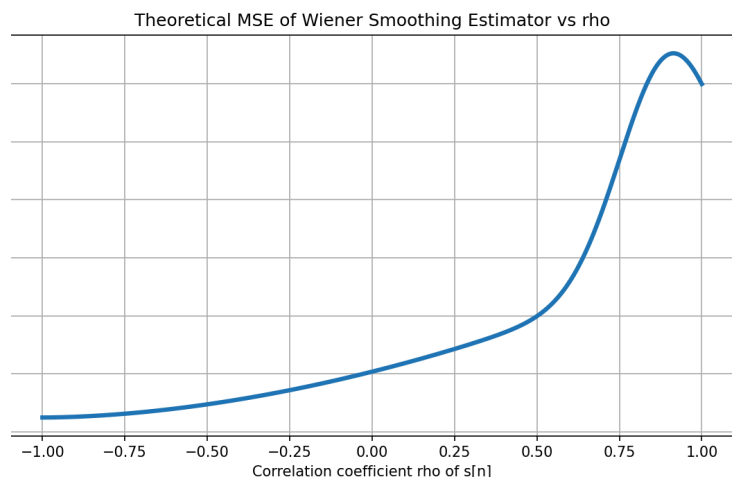
s is an AR(1) process with correlation coefficient ρ ;

w is an AR(1) process with ρ equal to 0.9. Therefore, we know that it has a narrowband PSD that simulates a low-pass behavior.

We want to estimate $s[N/2]$ using the Wiener smoothing filter. How will the plot of the MSE look with respect to the correlation coefficient ρ of the process $s[n]$?

One must reason based on the fact that the Wiener smoothing filter works better when the spectra of the process s and the noise w are different. Therefore, the MSE will have a minimum for ρ equal to

minus 0.9, that is when the process s has only high-frequency components which contrast with the low-frequency components of the noise. On the other hand, the MSE will reach a maximum when the ρ of the process s coincides with that of the process w . In this case the two signals are similar and the Wiener filter cannot estimate well, since both noise and signal have only low-frequency components.



3. How do we estimate the frequency F_0 in a sinusoidal/cosinusoidal process when it is not known: he just wanted to know that the peak frequency is found as the peak of the periodogram (ML estimator). Then, he asked to comment on the catastrophic errors that affect the estimate and to explain the MSE/CRB vs SNR graph of the estimate. He asked me which portion of the graph was due to the catastrophic errors and why it had a linear trend, such that the MSE increased linearly as the SNR decreased (he wanted to know that it is a sort of vicious cycle: as the SNR becomes worse, increasing the rate of catastrophic errors, that, by increasing, further worsen the SNR, and so on)
4. Describe in words the estimation of Phase amplitude and frequency with ML (all passages but in THIS CASE, didn't ask the demonstration, just to explain how by

words), describe the graph of pag 217 of chapter 4 (he opened the graph to let you comment it)

5. Describe the example and explain how can we approximate an ARMA model with an AR of higher order (the example was one in the ARMA PP and he asked to explain the different among various Graph). Explain why periodogram is not a consistent estimate in case of the continuous part of pdf